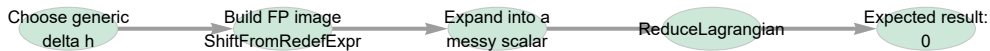


# Graviton Basis Toolkit Validation Notebook

This notebook wraps the reproducible reducer validation stored in `graviton_basis_toolkit_validation.wls`. It is intended as the notebook counterpart to the script-based test run.



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## Current Stored Results

The table below is loaded from `graviton_basis_toolkit_validation_output.wl`, which was produced by the most recent validation run.

| Sector | IBPCount | RedefRank | PhysicalCount | DeltaBasisLength | LagTermCount | Reduced to 0? |
|--------|----------|-----------|---------------|------------------|--------------|---------------|
| {3, 2} | 14       | 4         | 10            | 4                | 18           | True          |
| {3, 4} | 64       | 33        | 31            | 33               | 240          | True          |

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## How to Re-run

Evaluate the following input cell from the notebook front end or run the corresponding script directly from the kernel.

```
Get["graviton_basis_toolkit_validation.wls"]
```

The script writes `graviton_basis_toolkit_validation_output.wl` and uses `graviton_basis_validation_cache/` for sector caches.

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## What the Validation Proves

- It constructs Lagrangians that are in the image of a field redefinition of the Fierz-Pauli kinetic term.
- It expands those images so they do not transparently look like EOM times  $\delta h$ .
- It runs `ReduceLagrangian` on the expanded expressions.
- It checks that the reduced expression and all physical coordinates vanish.

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## Useful Follow-up

```
Get["graviton_basis_toolkit_validation_output.wl"]
```

You can inspect the returned association directly, or compare the stored image coefficients against the redefinition basis of the relevant sector.

